

UML Design Diagram

Project Title: Library Management System (LMS)

Project Type: Full-stack Web Development Capstone Project

Document Version: 1.0

Summary

Below shows the UML diagram for Library Management System using both mermaidchart codes and visual diagram

Feature Diagram

Login & Role Flow (Mermaid Flowchart)

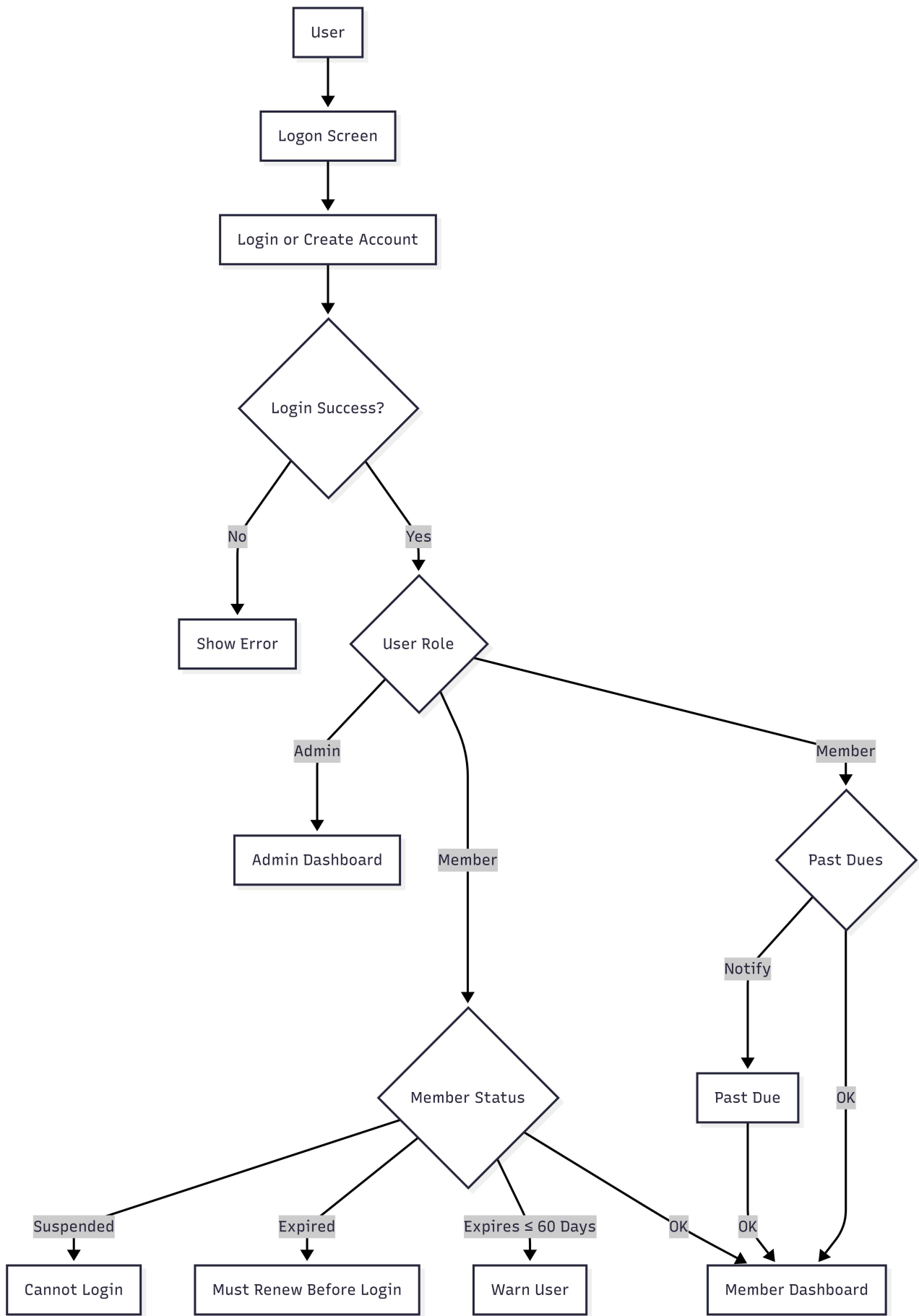
flowchart TD

```
A[User] --> B[Logon Screen]
B --> C[Login or Create Account]
C --> D{Login Success?}
D -->|No| E[Show Error]

D -->|Yes| F{User Role}
F -->|Admin| G[Admin Dashboard]

F -->|Member| H{Member Status}
F -->|Member| I{Past Dues}
I -->|Notify| J[Past Due]
I -->|OK| M[Member Dashboard]
J -->|OK| M[Member Dashboard]

H -->|Suspended| X[Cannot Login]
H -->|Expired| Y[Must Renew Before Login]
H -->|Expires ≤ 60 Days| Z[Warn User]
H -->|OK| M[Member Dashboard]
```



Member Dashboard Features

flowchart TD

M[Member Dashboard] → ML[My Loans]

M → MR[My Reservations]

ML → B1[Borrow Books]

B1 → B2{Borrow ≤ 3?}

B1 → B3{Fines < \$10?}

B1 → B4{Fine ≤ \$20/book?}

ML → MS[Manage Status]

MS → R1[Return Books]

MS → R2[Pay Fines]

MS → R3[Renew ≤ 2 Times]

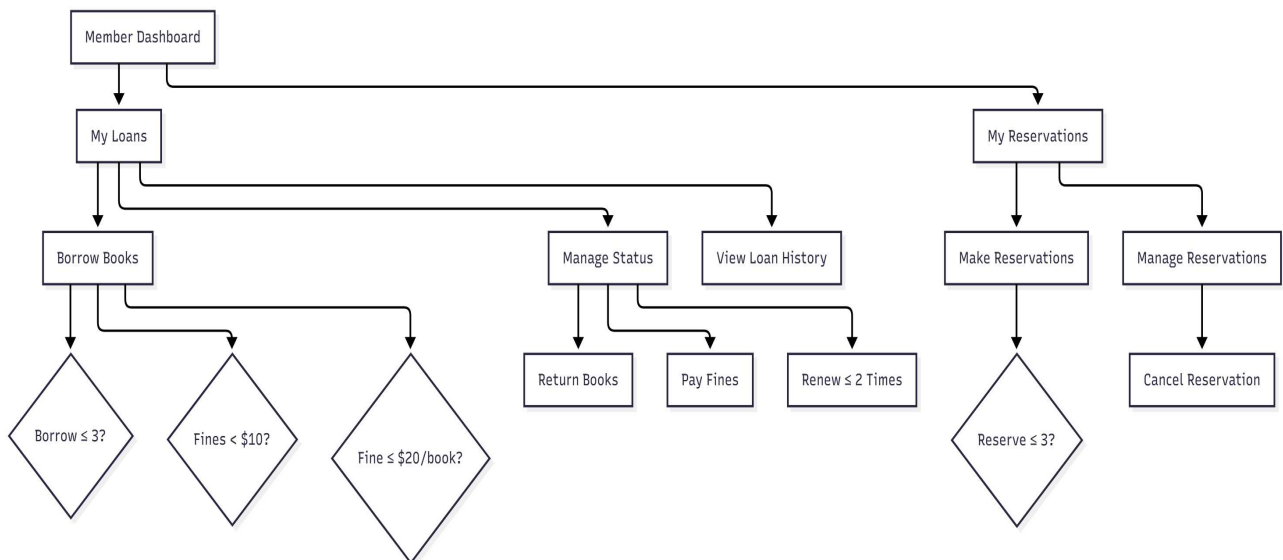
ML → VH[View Loan History]

MR → Rsv1[Make Reservations]

Rsv1 → Rsv2{Reserve ≤ 3?}

MR → RsvM[Manage Reservations]

RsvM → RsvC[Cancel Reservation]



Admin Dashboard Features

flowchart TD

G[Admin Dashboard] → U1[Manage Users - CRUD]

G → Bk1[Manage Books - CRUD]

G → AL[Active Loans]

AL → AL1[Renew on Behalf of Member]

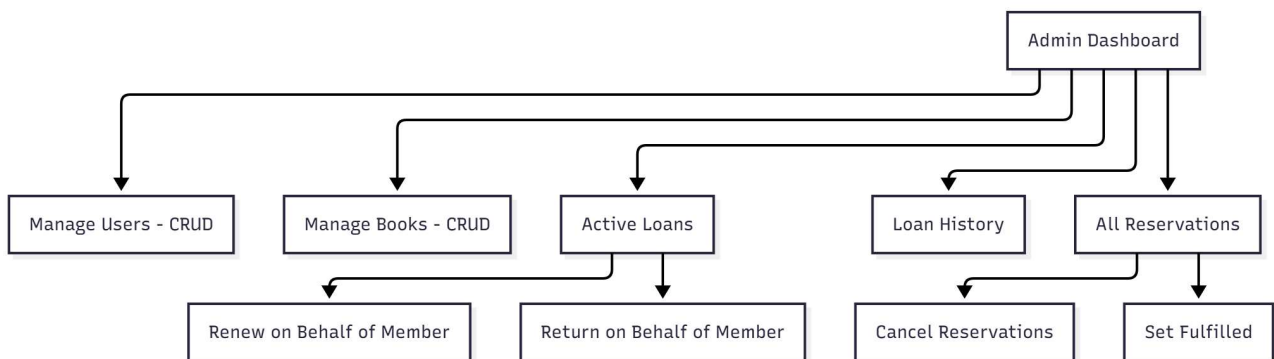
AL → AL2[Return on Behalf of Member]

G → LH[Loan History]

G → AR[All Reservations]

AR → ARC[Cancel Reservations]

AR → ARF[Set Fulfilled]

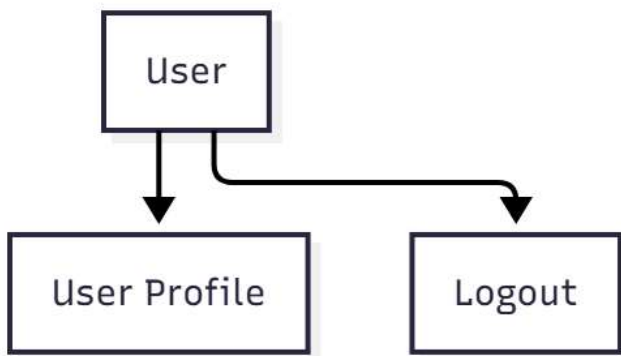


Common Functions (All Users)

flowchart TD

User → C1[User Profile]

User → C2[Logout]



Class Diagram

classDiagram

```
%% =====  
%%          USER CLASSES  
%% =====
```

```
class User {  
  +userId : String  
  +name : String  
  +email : String  
  +passwordHash : String  
  +role : UserRole  
  +status : MemberStatus  
  +membershipExpiry : Date  
  +login()  
  +updateProfile()  
}
```

```
class Admin {  
}
```

```
class Member {  
  +borrowCount : int  
  +reservationCount : int  
  +cumulativeFines : decimal  
  +canBorrow() bool  
  +canReserve() bool  
  +renewMembership()  
}
```

```
User <|-- Admin  
User <|-- Member
```

```
%% =====  
%%          ENUMS  
%% =====
```

```
class UserRole {  
  <<enumeration>>  
  ADMIN  
  MEMBER  
}
```

```
class MemberStatus {  
  <<enumeration>>  
  ACTIVE
```

```
SUSPENDED
EXPIRED
}
```

```
class BookStatus {
  <<enumeration>>
  AVAILABLE
  BORROWED
  RESERVED
}
```

```
class ReservationStatus {
  <<enumeration>>
  ACTIVE
  CANCELLED
  FULFILLED
}
```

```
%% =====
%%          BOOK CLASSES
%% =====
```

```
class Book {
  +bookId : String
  +title  : String
  +author : String
  +category : String
  +status : BookStatus
}
```

```
%% =====
%%          LOANS
%% =====
```

```
class Loan {
  +loanId : String
  +borrowDate : Date
  +dueDate : Date
  +returnDate : Date
  +renewCount : int
  +renew()
  +returnBook()
  +calculateFine()
}
```

```
Member --> Loan : "borrows 0..3"
Book --> Loan : "is loaned in 0..1"
```

```
%% =====  
%%      RESERVATIONS  
%% =====
```

```
class Reservation {  
  +reservationId : String  
  +reservationDate : Date  
  +status : ReservationStatus  
  +cancel()  
  +fulfill()  
}
```

Member --> Reservation : "makes 0..3"

Book --> Reservation : "is reserved by 0..*"

```
%% =====  
%%      FINES  
%% =====
```

```
class Fine {  
  +fineId : String  
  +amount : decimal  
  +reason : String  
  +dateIssued : Date  
  +paid : bool  
  +pay()  
}
```

Loan --> Fine : "generates 0..*"

Member --> Fine : "owes 0..*"

